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Experimental and numerical analysis of molten pool and keyhole profile during high-power deep-penetration laser welding

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ABSTRACT

High power laser welding is as an ideal advanced technology for joining the thick steel plates with high efficiency in the industries. This paper proposes a 3D numerical simulation modeling method to investigate keyhole profile characteristics and the effect of keyhole evolution on the molten pool, and understand weld formation process and defect generation, including the welding phenomena such as swelling, column and spatter in the weld pool during high-power deep-penetration laser welding. To reveal the weld formation process, the gas layer, keyhole free surface, molten pool surface, and solid-liquid-vapor three-phase transformation are considered in the model. Based on the proposed model, the keyhole periodic variation profiles and molten pool evolution are calculated. The interaction between the keyhole and molten pool dynamic behavior and weld defects are also identified and discussed in details. The numerical simulation results are compared with the experimental and literature results and good agreements are achieved. It is found that the constriction and bulge formed on the rear keyhole wall is the key factor causing periodical change of keyhole inlet. The oscillating interface between the keyhole and molten pool is often accompanied by swellings and columns, which is the main factor for spatter formation.

1. Introduction

High power laser welding has been as an ideal advanced technology for joining the thick steel plates with high efficiency and widely used in the industries. The ultrahigh peak power density from a 10 kW fiber laser can reach MW mm^{-2} levels [1], which enables the thick plates to be joined with single-pass welding and hence greatly improves the welding efficiency. The maximum weld depth with high power fiber laser welding has been over than 12 mm [2]. In the conventional method, the pre-processing for specimen and multi-pass welding are often required during the thick steel plates joining [3]. Since the high power laser welding process is affected by the complex heat and mass transfer behavior, the weld defects like undercut [4], spattering [1] and humping [5,6] are easily generated. Especially, the keyhole and molten pool dynamic behavior are becoming more and more instable as the increase in welding speed. The defects will emerge in the weld much more frequently. Improving the welding efficiency without sacrificing weld quality is severely limited by the weld defects. Therefore, understanding the formation process of instable keyhole profile and the dynamic characteristics of interface between the keyhole and surrounding molten pool in the high power welding is of great significance for improving the productivity and weld appearance quality.

To study the dynamic behavior of the keyhole and molten pool in the high power laser deep penetration welding, experimental efforts have been made to observe the welding process. Fabbro [7] defined the characteristic regimes of laser welding in a large welding speed range. The humping weld seam with strong undercuts was observed at above 20 m/min high speed welding in their study. Kawahito et al. [1] investigated the effects of laser power, power density and welding speed on the formation of sound weld in the bead-on-plate welding of type 304 stainless steel plates with 6 kW and 10 kW high power fiber laser. The welding phenomena recorded by high-speed video cameras and X-ray transmission real-time imaging system were clarified. They recognized that the humping formation was dependent upon the volume of molten metal above the surface, strong melt flow and other dynamic or static factors. Zhang et al. [8] presented a modified sandwich specimen with a 10 kW fiber laser used for observing the keyhole profile during the deep penetration laser welding. In their experiments, the distinct keyhole wall, moving liquid shelf on the front

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keyhole wall and accompanying hydrodynamic and vapor phenomena are directly observed. Li et al. [9] studied the dynamic behavior of keyhole based on a butt-joint configuration of transparent glass and stainless steel during the high-power deep-penetration welding. Their results showed that the gauffers and vapor whirlpool were the main factors causing the vapor-generated wave on the rear keyhole wall and fluctuation of keyhole profile, which resulted in the changes of keyhole inlet and molten pool accompanied by swellings, columns and spatter. They also investigated the relationship between spatter formation and molten pool dynamic behavior using the high-speed camera and X-ray transmission imaging system during the high power laser welding under different welding parameters [10]. Zhang et al. [11] observed the dynamic behavior of the keyhole, vapor plume, and weld pool with the spatters formation and analyzed the formation mechanisms of spatter in both top and bottom surfaces in high power fiber laser welding of thick plate. The spatter ejected from the keyhole inlet was mainly affected by the upward molten metal flow above the keyhole and strong shear stream of the directed vapor plume force at partial penetration. For the full penetration of the keyhole situation, the viscous friction drag associated with high-speed motion of vapor plume was the crucial driving force for spatter generation. Nakamura et al. [12] clarified the relationship among the laser-induced plume, molten metal flowing and spatters formation mechanisms during the 10 kW high power laser welding of pure titanium plate. Luo et al. [13] performed the welding experiments to investigate the coupling interaction mechanism between the molten pool and metallic vapor by analyzing their behavior from high-speed camera. The results showed that the fluctuation of molten pool surface was greatly affected by the metallic vapor, which plays major role in the weld bead formation. However, the application of experimental methods is limited due to the fact that the equipment is often costing and bulky and observing process is time and labor consuming. Moreover, the interfacial interaction between the molten weld pool and keyhole has been not understood well.

Computational fluid dynamics (CFD) methods have recently been discussed in the literature [14–17] and applied to high energy beam welding including the phenomena observed in experiments such as weld bead formation and defect generation. Panwisawas et al. [14] proposed a physics-based model considering the heat transfer, fluid flow and interfacial interactions to simulate keyhole and porosity formation during laser welding of Ti-6Al-4V titanium alloy. Pang et al. [15] developed a novel 3D transient multiphase model which took the self-consistent keyhole, metallic vapor plume and weld pool dynamic behavior into consideration in the deep penetration fiber laser welding. The obtained weld bead dimensions, transient keyhole instability, weld pool evolution, and vapor plume dynamic behavior were kept consistent with the experimental results. Tan et al. [16] developed a three-dimensional transient model to investigate the dynamic behavior of keyhole, vapor plume and molten pool. The complex surface phenomena on the keyhole wall and molten pool hydrodynamic behavior were calculated and validated against the experiments. Bachmann et al. [17] investigated the effect of alternating current magnetic field on the laser beam keyhole welding based on the three-dimensional turbulent steady state numerical model. Their results indicated that the gravity drop-out caused by the hydrostatic pressure could be prevented by alternating current magnetic field. However, the calculation of evolution of the weld pool and keyhole and the prediction for weld penetration and width in the normal welding are mainly focused by these methods. They may not be efficient for investigating the keyhole profile characteristics and molten pool dynamic behavior in the high power fiber laser welding. In particular, the formation processes of weld appearance defects which are affected by the molten pool dynamic behavior are not involved.

Only a few methods for the analysis of keyhole profile characteristics and the effect on the molten pool and weld defects formation in high power laser welding based on the numerical simulation have been reported in the previous literatures. Zhang et al. [18] simulated the coupling behavior between keyhole and molten pool in laser full penetration welding using 16 kW maximum power thin disk laser. The weld cross-section morphology and molten pool dynamic behavior were identified and the calculated results agreed well with the experimental results. The keyhole wall features and the interaction between the keyhole and molten pool, especially for the influences on the weld defect formation during the high power laser welding were still not revealed. Therefore, this paper proposes a 3D numerical simulation modeling method to investigate keyhole profile characteristics, the effect of keyhole evolution on the molten pool, and understand the weld formation and defect generation, including the welding phenomena such as swelling, column and spatter in the weld pool during high-power deep-penetration welding. The gas layer above the molten pool, keyhole free surface, molten pool free surface, and solid-liquid-vapor three-phase transformation are considered in the model to reveal the weld formation process. The keyhole periodic variation profiles, molten pool dynamic behavior, the interaction between the keyhole and molten pool behavior, and weld defects are analyzed based on the calculation results. The numerical simulation results show good agreements with the experimental and literature results.

The rest of the paper is summarized as follows: In Section 2, the experiments are designed and conducted. The description of the numerical simulation modeling including the boundary conditions and numerical implementations is provided in Section 3. The Section 4 introduces the model validation and comparison of the simulated and experimental results, keyhole profile characteristics, molten pool behavior, interfacial interaction of keyhole and molten pool. Finally, the conclusions of the current research are offered.

2. Experimental set-up

In this paper, the 8 mm thickness SUS316L stainless steel plate is selected as the base material. Table 1 shows the chemical composition. The whole laser welding process is conducted using a fiber laser (IPG YLS-10000) with the continuous wave. The maximum output power is 10 kW. The optical fiber diameter is 0.2 mm and the laser beam wavelength is 1070 nm. The output laser beam is converted into parallel light by the collimation lens with 150 mm focal length and then focused into spot diameter 0.4 mm at the focal point with lens of 300 mm focal length. The beam intensity distribution is approximate Gaussian distribution and the top surface of workpiece is set as the focal plane. The welding head is equipped with coaxial air blowing protection device and installed on a welding robot, KUKA. The laser welding system is shown in Fig. 1 and the corresponding technical parameters are tabulated in Table 2.

To obtain the keyhole and molten pool behavior, the welding process is observed by a high speed CMOS camera. The detailed keyhole and molten pool behavior are captured with 5000 frame per second (fps) from the top view. The 10 W diode laser with 808 nm wavelength is used to illuminate the welding zone. A narrow band-pass (808 ± 3 nm) interference filter is installed in front of the camera lens to improve the contrast between the molten pool and other region. To avoid the oil pollution and oxide film, all of the surfaces of specimens are brushed using fresh wire brush and cleaned by acetone swabbing carefully in advance. The shielding gas, Argon, is supplied laterally with a flow rate of 20 L/min.

Table 1
Chemical composition of the SUS316L stainless steel (Wt.%).

C	P	S	Si	Mn	Cr	Ni	N	Mo	Cu	Fe
0.021	0.039	0.001	0.77	1.019	16.92	12.16	0.033	2.03	0.20	Bal.

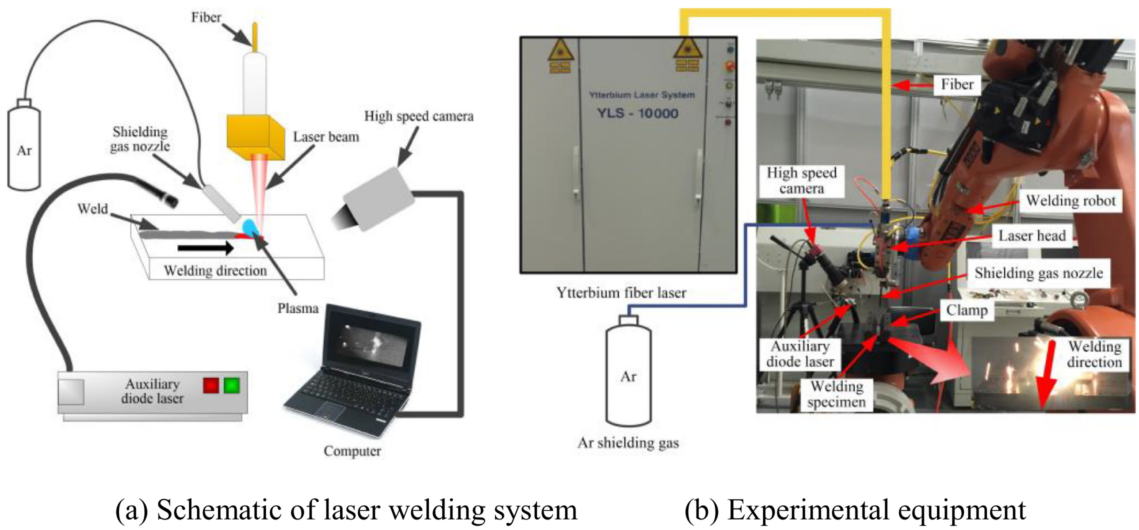

Fig. 1. Laser welding system.

Table 2
Technical parameters of laser welding system.

Parameters	Value
Laser	IPG YLS-10000
Maximum output power	10 kW
Beam wavelength (nm)	1070
Optical fiber diameter (mm)	0.2
Focal length of collimation lens (mm)	150
Focal length of focus lens (mm)	300
Focal point diameter (mm)	0.4

After welding, the prepared metallographic samples cut from the weld cross-section were ground with suitable sandpapers, polished using polishing agent and then etched for 30 s by keller's reagent. Under the optical microscope, the metallographic micrographs were obtained. The dimensions of weld bead were recorded based on the metallographic micrographs.

3. Numerical modeling

3.1. Governing equations

The schematic of high power laser welding is shown in Fig. 2. To make the model tractable, the liquid in the weld pool is assumed to be laminar, incompressible and Newtonian. The heat and mass transfer, and fluid flow in 3D Cartesian coordinate are described as follows [19,20].

Continuity equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{V}) = 0 \quad (1)$$

Momentum equation

$$\frac{\partial (\rho \vec{V})}{\partial t} + \nabla \cdot (\rho \vec{V} \vec{V}) = -\nabla P + \nabla \cdot (\mu \nabla \vec{V}) - \frac{\mu}{K} \vec{V} + F_M \quad (2)$$

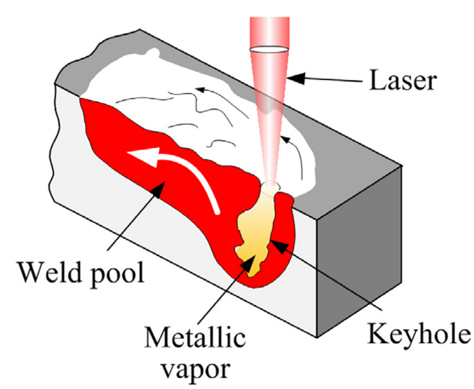

Fig. 2. Schematic of the high power laser welding.

Energy equation

$$\frac{\partial (\rho H)}{\partial t} + \nabla \cdot (\rho \vec{V} H) = \nabla \cdot (k \nabla T) + S_E \quad (3)$$

where ρ, t, $\vec{V}$, P, H, μ and k are the density, time, vector of velocity, pressure, enthalpy, viscosity and thermal conductivity, respectively. K is isotropic permeability. F_M is the source term in momentum equation and S_E is the source term of energy equation.

The mushy region between the solid and liquid phase is treated as porous media. The isotropic permeability from Carman-Kozeny equation can be calculated as [21]:

$$K = K_0 \frac{f^3 + \tau}{(1 - f)^2} \quad (4)$$

where K_0 is an empirical constant and τ is a small constant for convergence purpose. f is the liquid fraction which is defined by the enthalpy-porosity method [19] as:

$$f = \begin{cases} 0 & \text{if } T \leqslant T_s \\ \frac{T-T_s}{T_l-T_s} & \text{if } T_s < T < T_l \\ 1 & \text{if } T \geqslant T_l \end{cases} \tag{5}$$

where T_s and T_l are the solidus temperature and liquid temperature, respectively.

The enthalpy H is calculated as:

$$H = h_{ref} + \int_{T_{ref}}^{T} C_p dT + fL \tag{6}$$

where h_{ref} is reference enthalpy and T_{ref} is the reference temperature. C_p, T, L are specific heat, temperature of the workpiece and latent heat, respectively.

3.2. Boundary conditions

During mathematic modeling for high power deep penetration laser welding, the treatment for boundary conditions is complex and important due to the involved complicated physics phenomena. The detailed boundary conditions for the molten pool and keyhole dynamic behavior in the high power laser welding are described as follows.

3.2.1. Top, side and bottom surface

The top surface is treated as the pressure outlet in the numerical calculation. The free surface of molten pool is demonstrated in the interface between gas layer and workpiece in the calculation domain. The side and bottom surface are defined as the wall condition. The thermal boundary of wall including radiation and convection is described as [22]:

$$k\frac{\partial T}{\partial \vec{n}} = -\varepsilon\sigma(T^4 - T_0^4) - h_{conv}(T - T_0) \tag{7}$$

where T and T_0 are the workpiece temperature and ambient temperature, respectively. ε is the surface radiation emissivity, σ is the Stefan-Boltzmann constant and h_{conv} is the convection coefficient.

3.2.2. Keyhole and molten pool surface

The transient deformation of keyhole and weld pool free surface is tracked by the volume of fluid (VOF) model [23]. The VOF model is described as follows:

$$\frac{\partial F}{\partial t} + (\nabla \cdot \vec{V}F) = 0 \tag{8}$$

The phase interface in the whole domain can be calculated by Eq. (8). If $F = 0$, the cell is full of gas; if $F = 1$, the cell is full of liquid; if $0 < F < 1$, it denotes the cell is on the interface of liquid and gas.

For the keyhole free surface, the thermal boundary conditions considering the laser absorption, heat convection and radiation, and evaporation can be expressed as [19]:

$$k\frac{\partial T}{\partial \vec{n}} = q_{absorb} - \varepsilon\sigma(T_p^4 - T^4) - h_{conv}(T_{gas} - T) - W_{evap}H_v \tag{9}$$

where q_{absorb} is the absorbed laser energy. T_p is the average temperature of keyhole plasma and T_{gas} is the gas phase temperature. W_{evap} is the evaporation rate and H_v is the evaporation latent heat.

In the laser keyhole welding, the energy absorbed from the laser beam is described by the inverse Bremsstrahlung absorption above the keyhole and Fresnel absorption on the keyhole wall [24]. The Gaussian damped heat source is selected for modeling of the Fresnel absorption and the double ellipsoid heat source is adopted to express inverse Bremsstrahlung absorption. The absorbed laser energy q_{absorb} in the high power laser welding is calculated as follows [19]:

$$q_{absorb}(x,y,z) = q_f(x,y,z) + q_r(x,y,z) + q_g(x,y,z) \tag{10}$$

$$q_f(x,y,z) = \frac{6\sqrt{3}f_1 f_f Q}{a_f bc\pi\sqrt{\pi}} \exp\left(-3\left(\frac{(x+l-u_{ws}t)^2}{a_f^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2}\right)\right) \tag{11}$$

$$q_r(x,y,z) = \frac{6\sqrt{3}f_1 f_r Q}{a_r bc\pi\sqrt{\pi}} \exp\left(-3\left(\frac{(x+l-u_{ws}t)^2}{a_r^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2}\right)\right) \tag{12}$$

$$q_g(x,y,z) = \frac{6f_2 Q}{\pi r_0^2 H(1-e^{-3})} \frac{mz + r_0}{mH + 2r_0} \times \exp\left(\frac{-3((x+l-u_{ws}t)^2 + y^2)}{r_0^2}\right) \tag{13}$$

$$Q = \eta P_{laser} \tag{14}$$

where $q_f(x,y,z)$ and $q_r(x,y,z)$ are the power densities of the double ellipsoid heat source in the front and rear quadrant parts, respectively. $q_g(x,y,z)$ is the power density in the Gaussian damped heat source. Q is the effective absorbed energy. f_f and f_r are the distribution factors of the heat deposited in front and rear quadrant of ellipsoid. f_1 and f_2 are the energy distribution coefficients for the double ellipsoid heat source and Gaussian heat source. a_f, a_r, b and c are parameters to define the size and shape of ellipsoid to express the heat source distribution. l is a positive constant to set start welding point. u_{ws} is the welding speed. r_0 is the radius of Gaussian heat source and m is the damping coefficient. H is the heat source penetration. η is the absorption coefficient and P_{laser} is the laser welding power [25].

For the free surface of molten pool, the thermal boundary condition is expressed as the following formula [22]:

$$k\frac{\partial T}{\partial \vec{n}} = -\varepsilon\sigma(T^4 - T_0^4) - h_{conv}(T - T_0) \tag{15}$$

3.2.3. Driving forces

During the simulation, the surface tension, recoil pressure, hydrostatic pressure, and hydrodynamic pressure are considered in the model. The pressure on the keyhole surface is expressed as [26]:

$$P = P_\gamma + P_r + P_g + P_h \tag{16}$$

where P_γ, P_r, P_g and P_h are the surface tension, recoil pressure, hydrostatic pressure and hydrodynamic pressure, respectively.

The surface tension is calculated as [23,27,28]:

$$P_\gamma = \gamma\kappa \tag{17}$$

$$\gamma = \gamma_0 - A(T - T_l) - R_g T\Gamma_s \ln(1 + K_s\alpha_s) \tag{18}$$

$$\frac{\partial\gamma}{\partial T} = -A - R_g\Gamma_s \ln(1 + K_s\alpha_s) - \frac{K_s\alpha_s\Delta H_0\Gamma_s}{T(1 + K_s\alpha_s)} \tag{19}$$

$$K_s = k_l \exp\left(\frac{-\Delta H_0}{R_g T}\right) \tag{20}$$

$$\kappa = -\left[\nabla\cdot\left(\frac{\vec{n}}{|\vec{n}|}\right)\right] = \frac{1}{|\vec{n}|}\left[\left(\frac{\vec{n}}{|\vec{n}|}\cdot\nabla\right)|\vec{n}| - (\nabla\cdot\vec{n})\right] \tag{21}$$

where γ is the surface tension coefficient and κ is the free surface curvature. γ_0 and $\boldsymbol{A}$ are the surface tension and temperature coefficient of pure iron. Γ_s, α_s, ΔH_0 and k_l are saturation parameter, sulfur thermodynamic activity, standard heat of absorption and entropy factor, respectively. $\vec{n}$ is the unit normal vector of free surface.

The recoil pressure P_r induced by evaporation process is described as [26,29]:

$$P_r = AB_0/\sqrt{T_w}\exp(-m_a H_v/N_a k_b T_w) \tag{22}$$

where A is the coefficient and B_0 is the evaporation constant. m_a is the atomic mass and H_v is the evaporation latent heat. N_a is the Avogadro's number and k_b is the Boltzmann constant. T_w is the temperature of keyhole wall.

Hydrostatic pressure P_g and hydrodynamic pressure P_h are expressed as the following formula [19]:

$$P_g = gh \tag{23}$$

$$P_h = \frac{u^2}{2g} \tag{24}$$

where g is gravitational acceleration, h is the height of the fluid in the weld pool and u is the velocity of the fluid.

For the free surface of molten pool, the pressure can be calculated as follows:

$$P_\gamma = \gamma\kappa \tag{25}$$

The buoyancy force induced by density gradient is calculated based on the Boussinesq approximation as [30]:

$$S_b = \rho g \beta (T - T_l) \tag{26}$$

where β is the thermal expansion coefficient. T_l is the liquidus temperature.

According to the set boundary conditions above, the recoil pressure and buoyancy force are treated as the source term of momentum equation, F_M and the energy source term, S_E, is defined based on the thermal boundary condition [31,32].

3.3. Numerical implementations

In the current study, the numerical model is solved by the finite volume method. The numerical calculation is conducted by the commercial CFD software, Fluent 17.0. The source terms in governing equations and boundary conditions are achieved by the user defined functions (UDF) written in C language.

The thermophysical material properties are listed in Table 3, which is determined by the commercial thermal parameter calculation software, JmatPro and other published literatures. During the numerical simulation, the small part of welding specimens with the dimension 10 mm × 3 mm × 12 mm is used in the calculation and the total number of cells is 100 × 30 × 120. The mesh model is divided into two layers. Since the SUS316L stainless steel with 8 mm thickness is used in the welding experiment, the bottom layer of mesh model with 8 mm thickness is treated as the base material SUS316L and the top one is set as the gas layer. The mesh model is as shown in Fig. 3.

Table 3
Thermophysical material properties in the numerical calculation [15,19,25].

Name	Symbol	Value	Units
Density of liquid phase	ρ	7200	kg m^{-3}
Solidus temperature	T_s	1697	K
Liquidus temperature	T_l	1727	K
Liquid Specific heat	C_p	780	J kg^{-1} K^{-1}
Thermal conductivity	k	35	W m^{-1} K^{-1}
Dynamic viscosity	μ	0.006	kg m^{-1} s^{-1}
Convective heat transfer coefficient	h_{conv}	80	W m^{-2} K^{-1}
Surface radiation emissivity	ε	0.4	
Stefan-Boltzmann constant	σ	5.67×10^8	W m^{-2} K^{-4}
Melting latent heat	L_m	2.74×10^5	J kg^{-1}
Evaporation latent heat	H_v	6.46×10^6	J kg^{-1}
Avogadro's number	N_a	6.02×10^{23}	
Effective radius of heat source	r_0	0.2	mm
Laser absorptivity of the material	η	0.7	
Thermal expansion	β	4.95×10^{-5}	K^{-1}
Evaporating temperature	T_{lg}	3270	K

4. Results and discussion

4.1. Model validation

The keyhole and weld pool for the high power laser welding with 9.0 kW laser power and 3 m/min welding speed are calculated based on the numerical model. The 3D transient keyhole profiles and molten pool dynamics are shown in Fig. 4. From the figure, the keyhole wall is not smooth and demonstrates irregular bulges. The temperature distribution on the front wall is found higher than other regions, which is due to the fact that the evaporation process is mainly in the front part as the laser beam moving forwards. These phenomena are kept consistent with the previous research results, such as Pang et al. [15] and Kaplan et al. [33]. The molten metal is flowing from the rear edge of keyhole to the rear of weld pool with fluctuations, which agrees well with the reported experimental results [17]. The keyhole and molten pool dynamic behavior are also observed from the high speed camera as shown in Fig. 5. It is seen that the calculated results are in good agreement with experiments.

The comparison of the weld dimensions and appearance from the simulated and experimental results is shown in Fig. 6. From Fig. 6(a), both of the depth and width of weld are in good agreement. From the top view, the simulated appearance also agrees well with the results from the experimental result. Therefore, the validity of the model is confirmed and the proposed model can be used for investigating the keyhole profiles and weld pool dynamics in the high power fiber laser welding.

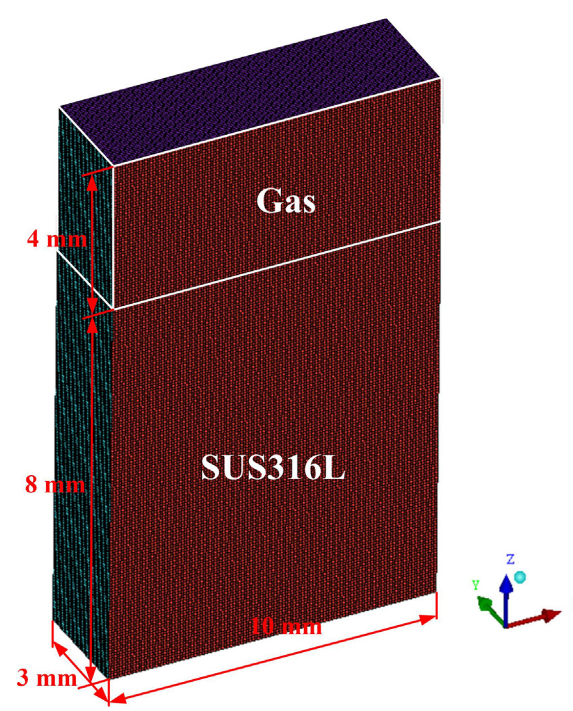

Fig. 3. The mesh model.

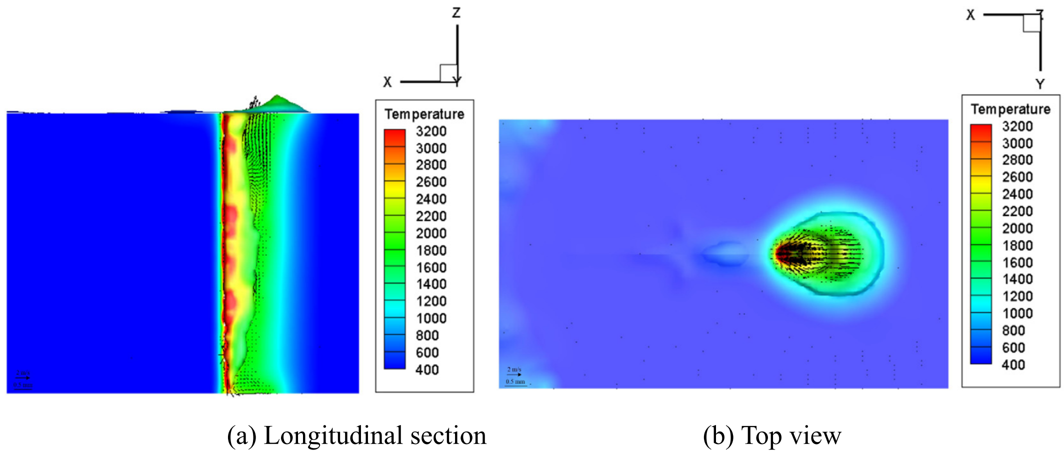

Fig. 4. The calculated keyhole and molten pool.

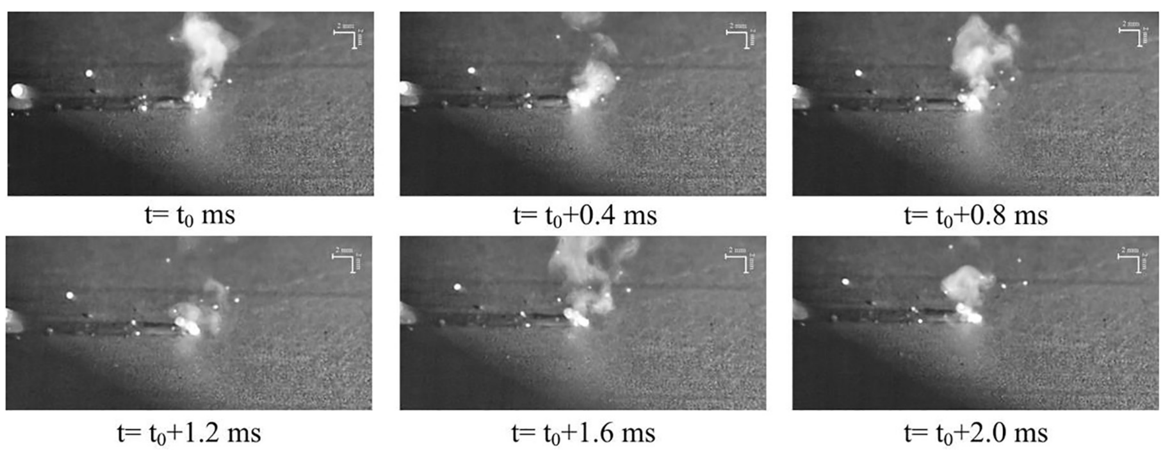

Fig. 5. High speed camera observed images of the keyhole and molten pool dynamic behavior.

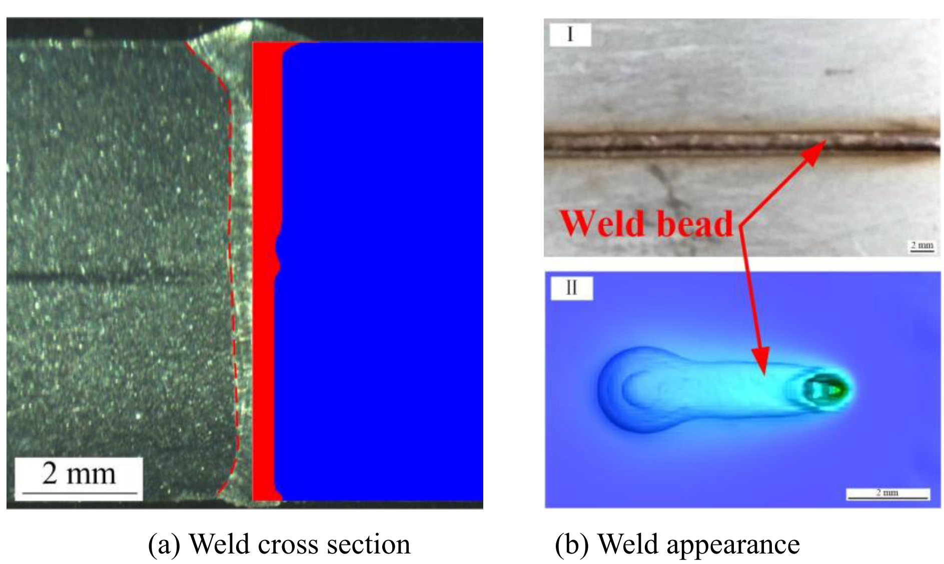

Fig. 6. The comparison of simulated and experimental results.

4.2. Keyhole profile characteristics

The calculated keyhole profile evolution in the longitudinal section during the high power fiber laser welding is shown in Fig. 7. The molten metal in the weld pool irradiated by the laser beam is evaporated rapidly and the keyhole depth increases to achieve the maximum depth, as shown in Fig. 7(a). The keyhole wall is rough with the constriction and bulge formed in the rear keyhole edge and the small gauffer emerges on the front keyhole wall. With the laser beam moving forwards, the front keyhole wall is irradiated by the laser beam directly, which causes the intense evaporation in the local region [10,34]. The formed gauffer on the front keyhole wall is moving and growing up. The intense evaporated vapor ejects from the direction perpendicular to the front edge of keyhole towards the rear keyhole wall, which enhances the local bulges formation, as shown in Fig. 7(b). The front keyhole wall is almost vertical during the welding process and the rear keyhole wall demonstrates dynamic variation during the moving process. Several constrictions and bulges are observed on the rear keyhole wall, as shown in Fig. 7(a)–(c). It is clearly seen that the formed constriction is moving upwards with the bulge growth which is induced by the recoil pressure, surface tension and

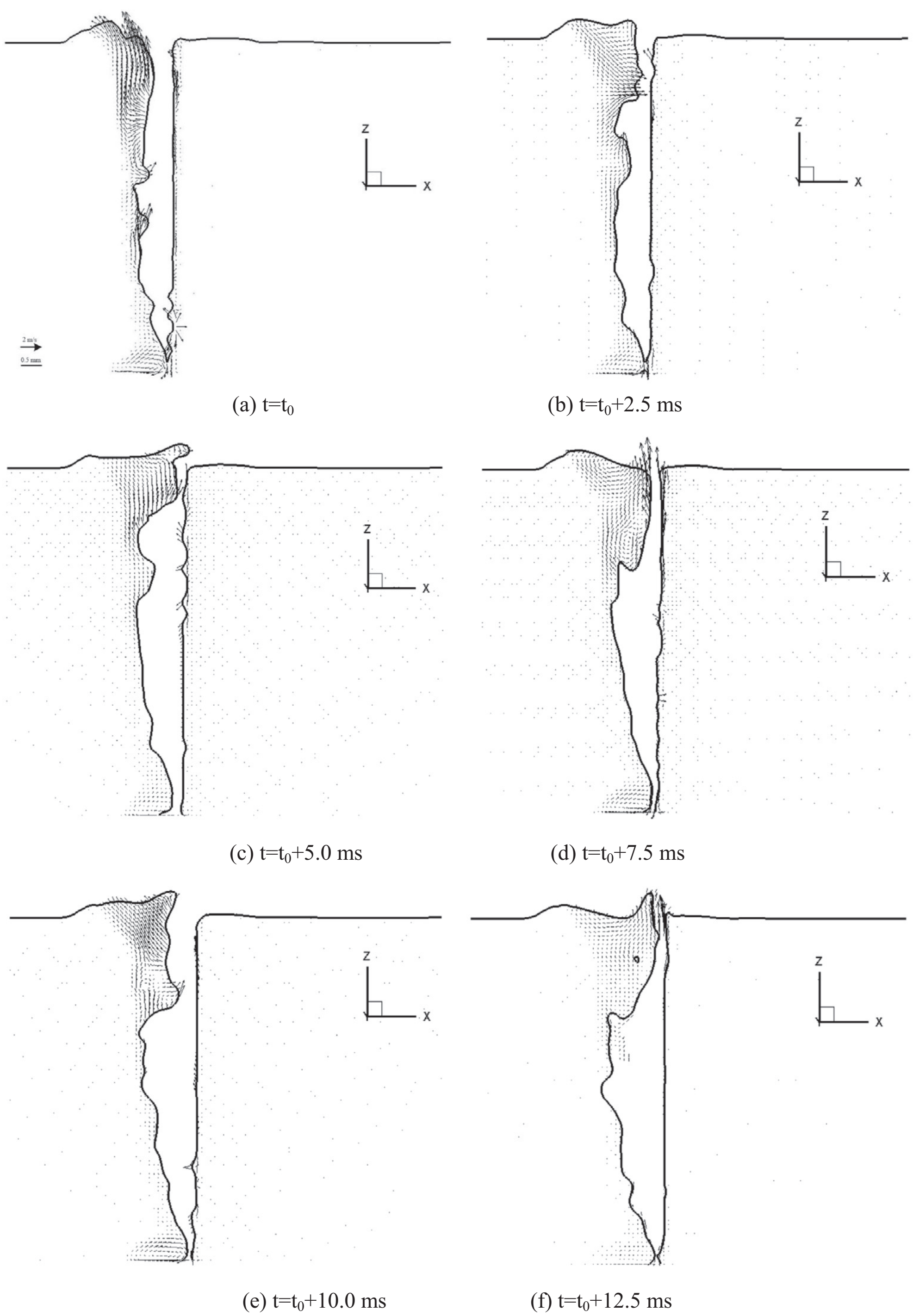

Fig. 7. The evolution of the keyhole profile in the high power fiber laser welding.

hydrostatic pressure. When the constriction moves to the keyhole inlet, the upwelling molten metal is accumulated near the rear keyhole wall and forms as the swelling, as shown in Fig. 7(b). After the bulge moves to the top surface, the diameter of the keyhole is reduced rapidly and strong vapor flows through the narrow keyhole inlet, which causes the column formation on the rear keyhole wall, as shown in Fig. 7(c) and (f). After the bulges near the top surface move out of the keyhole, the keyhole inlet recovers to the large mouth again. Therefore, the inlet of keyhole is kept in opening and closing dynamic variation, which is consistent with other observed experimental results [14,35]. The keyhole profile is periodicity oscillation with the irregular waves on the wall, as shown in Fig. 7(d)–(f). The constriction and bulge formed on the rear keyhole wall is the key factor causing periodical change of keyhole inlet.

4.3. The molten pool behavior

Fig. 8 shows the molten pool behavior in the high power fiber laser welding. In the initial stage, the molten metal is pushed from the rear edge of keyhole to the rear part of molten pool due to the influence of strong ejecting vapor plume. The liquid metal is accumulated at the rear side of molten pool as the hump, as shown in Fig. 8(a). With the increase of displaced molten metal, the height of molten pool is obviously growing up. There is a metal column formed inclined to the keyhole, which reduces the keyhole inlet diameter, as shown in Fig. 8(b). Sequentially, the stronger metallic vapor will be generated and pushes more molten metal to the rear part of molten pool. The molten pool surface is becoming fluctuation and demonstrates as uphill and downhill, as shown in Fig. 8(c). After that, a new swelling is formed behind the keyhole and the generated bulge on the rear keyhole wall encounters with the formed swelling under the effect of the strong vapor plume, as shown in Fig. 8(d) and (e). Simultaneously, the bulge grows along the vertical direction and forms a tall metal column with vapor plume, as shown in Fig. 8(e) and (f). The fluctuation wave is emerging in the molten pool. As the molten metal cooling down, the weld is formed in the rear part of molten pool after solidification. It is clearly demonstrated that the formation of weld bead surface is greatly affected by the molten pool dynamic behavior.

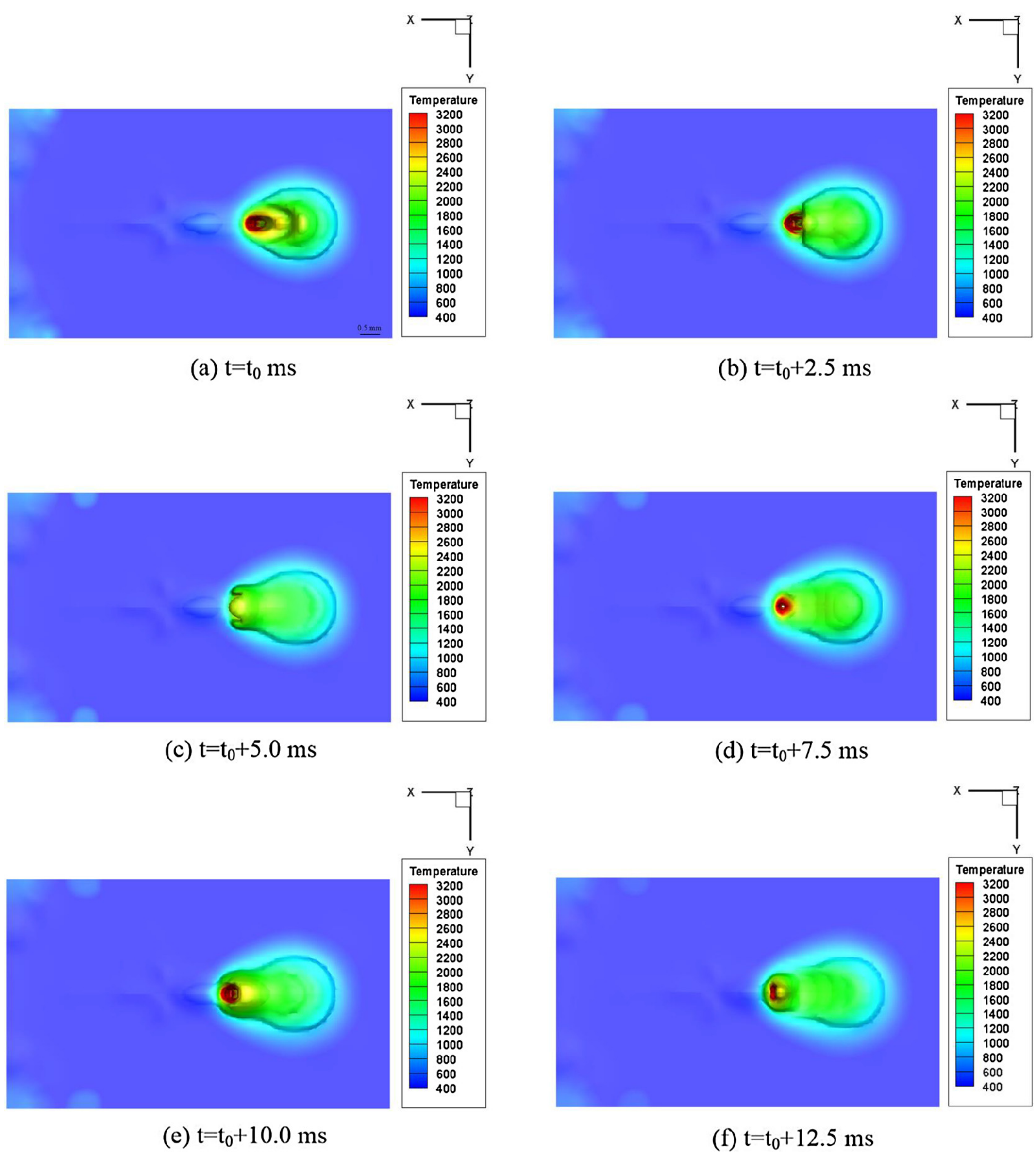

Fig. 8. The molten pool dynamic behavior in high power fiber laser welding.

4.4. Dynamic interaction between the keyhole and molten pool

The dynamic interaction of the keyhole and molten pool is shown in Fig. 9. The molten pool dynamic behavior is highly associated with the evolution of keyhole. As shown in Fig. 9(a), the keyhole is formed in the initial stage and the fluid metal flows from the rear wall of keyhole to the molten pool driven by the recoil pressure and surface tension. Since the small constriction and bulge are formed on the keyhole wall in this stage, there is no region with large pressure and the fluid flow in the molten pool is kept stable. As times goes, the keyhole affected by the metallic vapor is becoming instable and the large gauffers are formed on the rear wall of the

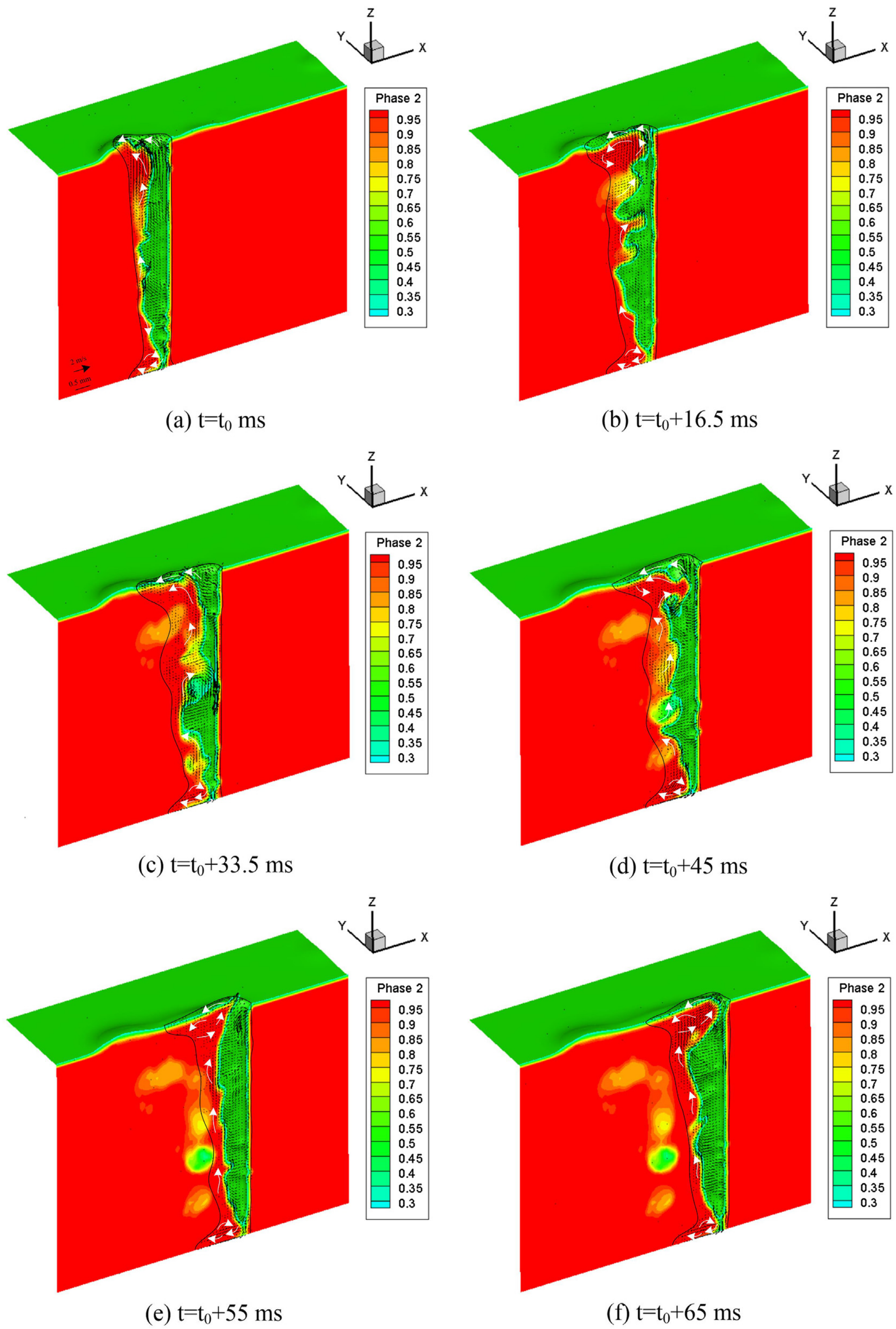

Fig. 9. The dynamic interaction between the keyhole molten pool.

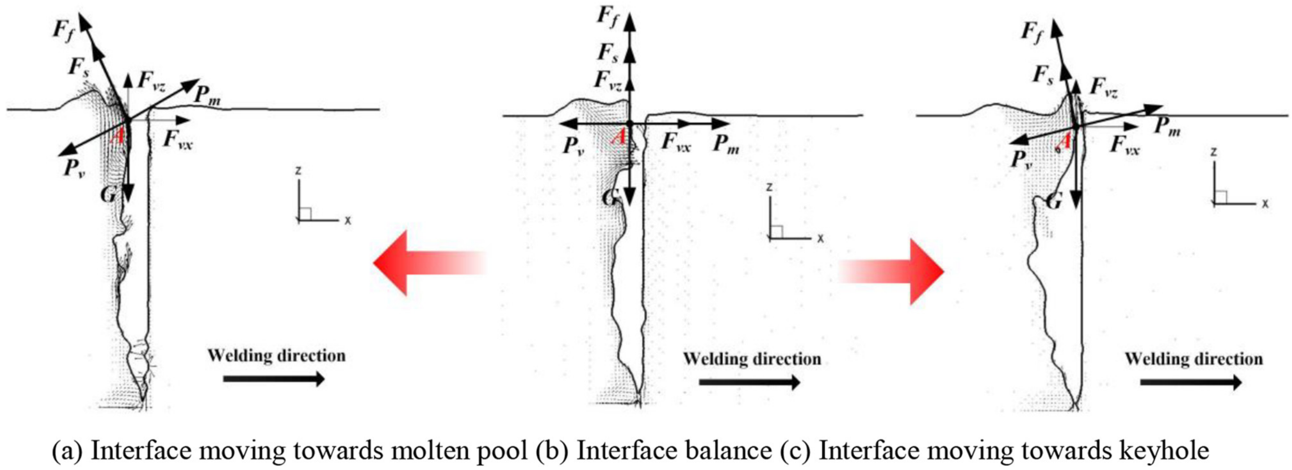

Fig. 10. Force analysis at the interface between the rear keyhole wall and molten pool surface.

keyhole from the bottom to top region, as shown in Fig. 9(b). Sequentially, the molten pool surface near the rear wall of the keyhole is humped as the swelling and the fluctuations in the molten pool surface are formed. The constriction is moving from the bottom to the keyhole inlet. After the constriction moving out, the molten metal is flowing to the rear part of molten pool. There is a downhill formed in the front part, as shown in Fig. 9(c) and (d). When several bulge regions on the keyhole wall move to the keyhole inlet, the molten pool interface is pushed to the keyhole side, which results in the rapid decrease of keyhole inlet. The column and swelling formed along the rear wall of the keyhole provides the condition for the spatter defect formation. Simultaneously, the wave is generated on the surface of molten pool, and the fluctuation of molten pool surface is formed, as shown in Fig. 9(e) and (f). This phenomenon is kept consistent with the recent reported experimental results of weld pool and keyhole dynamics characteristics in laser welding of Luo et al. [13]. The results show that the molten pool stability is greatly depended on keyhole dynamic behavior and hence affects the formation of weld bead surface.

The interaction of the keyhole and molten pool is mainly taken place at the interface between the molten pool and keyhole. The corresponding interaction mechanism can be revealed by the force analysis of the interface. Fig. 10 shows the force analysis of point A at the interface between the rear keyhole wall and molten pool. The keyhole stability can be evaluated by the opening and closing alteration of inlet. In the numerical modeling, the vapor recoil pressure which is normal to the keyhole wall surface is the main driving force keeping the keyhole open, while the surface tension, hydrostatic pressure and hydrodynamic pressure induced by the molten metal gravity and flowing prevent the keyhole formation. In the formation primary stage, the keyhole is stable as shown in Fig. 10(b), the force at point A on the keyhole wall is equilibrium, which can be expressed in horizontal and vertical directions as [13]:

$$P_m + F_{vx} = P_v \tag{27}$$

$$F_f + F_s + F_{vz} = G \tag{28}$$

where P_m is the hydrostatic pressure. F_{vx} and F_{vz} are the components of hydrodynamic pressure F_v in the x and z axis directions, respectively. P_v is the vapor recoil pressure. F_f is the friction force between the metallic vapor and molten metal. F_s is the surface tension and G is the gravity.

Since the vapor recoil pressure is highly depended by the temperature of keyhole wall, the force balance at the interface is easily broken by the instability of recoil pressure. When the recoil pressure P_v increases rapidly, the force at the keyhole side is over than that of the molten pool side. Consequently, the interface is pushed towards the molten pool side from the keyhole wall, which results in the uphill formation on the top surface of molten pool near the rear edge of keyhole. The corresponding dynamic behavior of the keyhole and molten pool are shown in Fig. 10(a). The relationship of the force at point A of the interface in both vertical and horizontal directions can be described as [13]:

$$P_{mx} + F_{vx} < F_{fx} + F_{sx} + P_{vx} \tag{29}$$

$$P_{mz} + F_{fz} + F_{sz} + F_{vz} < G + P_{vz} \tag{30}$$

where P_{mx} and P_{mz} are the components of hydrostatic pressure P_m in the x and z axis directions. F_{fx} and F_{fz} are the components of friction force F_f in the x and z axis directions. F_{sx} and F_{sz} are the components of surface tension F_s in the x and z axis directions, and P_{vx} and P_{vz} are the components of recoil P_v in the x and z axis directions.

When the recoil pressure P_v cannot keep the keyhole open and the molten metal near the rear wall of keyhole is pushed towards the keyhole due to the influence of hydrostatic pressure, hydrodynamic pressure and surface tension, the keyhole diameter will decrease and the swelling and column will be formed along the metallic vapor ejected from the keyhole, shown in Fig. 10(c). The force at point A on the keyhole wall can be expressed as [13]:

$$P_{vx} + F_{fx} + F_{sx} < P_{mx} + F v_x \tag{31}$$

$$G + P_{vz} < P_{mz} + F_{fz} + F_{sz} + F_{vz} \tag{32}$$

5. Conclusion

The keyhole characteristics and molten pool dynamic behavior during the high-power deep-penetration fiber laser welding have been investigated in this paper. The specific conclusions can be drawn as follows:

(1) The good agreement between simulation and experimental results demonstrates that the proposed 3D model is reasonable for studying the high power fiber laser welding process.
(2) The local bulge formation on the rear wall is caused by the intense evaporation on the front wall of the keyhole. The formed constriction moves upward to the top surface of keyhole with the bulge growth driven by the recoil pressure, surface tension and hydrostatic and hydrodynamic pressure, which results in the keyhole instability and periodic variation.

(3) The molten pool surface is frequently fluctuated and the swelling and column are often formed near the rear side of keyhole due to the effect of the keyhole profile oscillation, which provides the condition for spatter defect formation.
(4) The interaction mechanism between the keyhole and molten pool dynamic behavior has been revealed through the force analysis at the interface. When the force balance is broken, the interface between the keyhole and molten pool moves forwards or backwards, which causes the molten pool fluctuation and keyhole profile oscillation.

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Conflict of interests

None declared.

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